

Specialist Mathematics · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

Exam 1 ·

38 marks

reading 15 min · writing 60 min

Exam 2 ·

82 marks

reading 15 min · writing 120 min

State average on these questions

58 / 120 (48%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the hardest questions.

Sources

- [2023 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)

Exam 1 · technology free (38 marks, reading 15 min, writing 60 min)

Questions (technology free)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (3 marks)

2023 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 4\)](#) ↗

Find $b > 0$ given $\arg((b - i)^3) = -\frac{\pi}{2}$

Find b (3 marks)

Question 2 (3 marks)

2023 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

Velocity given as a function of position, $v = \frac{3x+2}{2x-1}$: acceleration at a point, limiting velocity

a. Acceleration when $x = 2$ (2 marks)

b. Limiting velocity as x grows (1 mark)

Question 3 (4 marks)

2023 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 8\)](#) ↗

Prove by induction a formula for the n th derivative of xe^{2x}

Induction proof (4 marks)

Question 4 (6 marks)

2023 VCAA Exam 1 · Question 10 · [open in the official exam PDF \(p. 10\)](#) ↗

Particle moving on a circle: double-angle rewrite, Cartesian path, a time from distance travelled, when position $\perp$ velocity

a. Write $5 - 6 \sin^2 t$ as $\alpha + \beta \cos(2t)$ (1 mark)

b. Show the Cartesian equation of the path (2 marks)

c. Find a from the distance travelled (1 mark)

d. All t with $\underline{\underline{r}}(t) \perp \underline{\underline{\dot{r}}}(t)$ (2 marks)

Question 5 (4 marks)

2024 VCAA Exam 1 · Question 9 · [open in the official exam PDF \(p. 12\)](#) ↗

Car with v^2 given in terms of position via $\arccos$: speed camera trigger, acceleration at $x = 12$

a. Is the device activated? (1 mark)

b. Acceleration at $x = 12$ (3 marks)

Question 6 (3 marks)

2024 VCAA Exam 1 · Question 10 · [open in the official exam PDF \(p. 13\)](#) ↗

Shortest distance between two skew lines with a parameter m

Values of m (3 marks)

Question 7 (5 marks)

2025 VCAA Exam 1 · Question 4 · [open in the official exam PDF \(p. 5\)](#) ↗

Waiting-time pdf: show the mean is $1/2$ using partial fractions; probability for a sample mean of 25

a. Show $E(T) = \frac{1}{2}$ (3 marks)

b. $\Pr(0.44 < \bar{T} < 0.5)$ (2 marks)

Question 8 (4 marks)

2025 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 6\)](#) ↗

Two particles collide at $t = 1$: find c , then a from perpendicular velocities, then a and b from equal acceleration magnitudes

a. Show $c = -4$ (1 mark)

b. Two possible values of a (2 marks)

c. Values of a and b (1 mark)

Question 9 (6 marks)

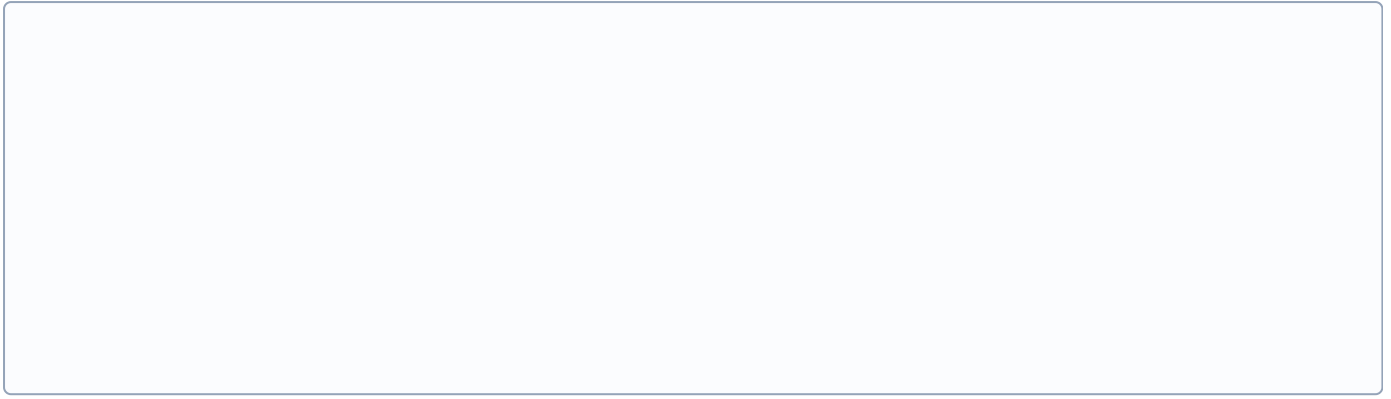
2025 VCAA Exam 1 · Question 9 · [open in the official exam PDF \(p. 10\)](#) ↗

Rational function with a removable discontinuity: rewrite as $-x - 1 + \frac{1}{x+1}$, make a hybrid continuous, sketch with a hole

a. Show $f(x) = -x - 1 + \frac{1}{x+1}$ (2 marks)

b. k for continuity at $x = 1$ (1 mark)

c. Sketch, labelling asymptotes (3 marks)



Exam 2 · CAS and a bound reference allowed (82 marks, reading 15 min, writing 120 min)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–E. One mark each.

Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 3\)](#) ↗

Given $|\bar{z}| = 4$, $\arg(z^3) = -\pi$ and z in the first quadrant, identify z^2

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 5\)](#) ↗

Mixing problem in a spa pool with unequal inflow and outflow rates

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 6\)](#) ↗

Reduction formula for $I_n = \int_0^1 (1-x)^n e^x dx$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 6\)](#) ↗

Integral for the surface area when part of $y = \cos^{-1} x$ is rotated about the y-axis

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 7\)](#) ↗

Time for a phone dropped from a rising balloon to hit the ground

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 7\)](#) ↗

$|\tilde{c} \cdot \tilde{n}|$ for a unit vector $\tilde{n}$ perpendicular to two given vectors

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 7 · 1 mark

2023 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 7\)](#) ↗

Magnitude of the difference of two unit vectors whose sum is a unit vector

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 8 · 1 mark

2023 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 7\)](#) ↗

Total vertical distance travelled by a thrown ball before landing

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 9 · 1 mark2024 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗Which statement is true for a hybrid function $\frac{x^2+3x-10}{x-2}$ with a defined value at $x = 2$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2024 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗ $\cos \frac{x}{2}$ in terms of $a = \sin x$ for x in the fourth quadrantYour answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2024 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 4\)](#) ↗

Integral for the length of one arch of a cycloid-type curve

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 6\)](#) ↗Scalar resolute of s on r , given the vector resolute of r on s Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2024 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 6\)](#) ↗Describing the motion $\tilde{r} = \sin t \tilde{i} + \cos(2t) \tilde{j}$ along a parabolaYour answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2024 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 7\)](#) ↗

Number of times one particle's velocity is perpendicular to another's position

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2024 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 8\)](#) ↗

Probability the edible flesh of four avocados totals more than a value

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2025 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗Counterexample to "if $f''(0) = 0$ then there is an inflection point at 0"Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2025 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 6\)](#) ↗Surface area integral for $x = kt$, $y = e^{kt}$ rotated about the x -axisYour answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2025 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 7\)](#) ↗

Velocity at the midpoint under constant acceleration

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark2025 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 8\)](#) ↗Angle between vectors when $\vec{a} \cdot \vec{b} = |\vec{a} \times \vec{b}|$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark2025 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 9\)](#) ↗

Unknown constants from the intersection point of two lines

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Extended-response questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (10 marks)2023 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 12\)](#) ↗

Seventh roots of unity: listing and plotting them, a ray on the Argand diagram, factorising $z^7 - 1$, and showing $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$

a. Verify $w = \text{cis } \frac{2\pi}{7}$ is a root (1 mark)**b.** Other roots in polar form (1 mark)**c.** Plot all seven roots (2 marks)**d.i.** Sketch the ray from 1 through w (1 mark)**d.ii.** Equation of the ray (1 mark)

e. Verify the factorised form of $z^7 - 1$ (1 mark)

f.i. Write $\text{cis } \frac{2\pi}{7} + \text{cis } \frac{12\pi}{7}$ as $A \cos(B\pi)$ (1 mark)

f.ii. Show the cosine sum is $-1/2$ (2 marks)

Question 2 (11 marks)

2023 VCAA Exam 2 · Section B · Question 5 · [open in the official exam PDF \(p. 20\)](#) ↗

Points, lines and planes: triangle area, distance from a point to a line, angle between a line and a plane, a normal line, distances and an intersection point

a. Vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and show the area is 1.5 (2 marks)

b. Shortest distance from B to AC (2 marks)

c. Acute angle between the line and plane ψ (2 marks)

d. Parametric equation of L (1 mark)

e. Shortest distance from the origin to ψ (2 marks)

f. Coordinates of D (2 marks)

Question 3 (10 marks)

2024 VCAA Exam 2 · Section B · Question 1 · [open in the official exam PDF \(p. 10\)](#) ↗

$f(x) = \frac{x^4 - x^2 + 1}{1 - x^2}$: sketch, volume about the y-axis, then a family $\frac{x^4 + b}{1 - x^2}$ and how many stationary points it has

a. Sketch with asymptotes and stationary points (3 marks)

b.i. Integral in y for the volume (2 marks)

b.ii. Volume to 1 d.p. (1 mark)

c. b for no asymptotes (1 mark)

d.i. b for exactly one stationary point (1 mark)

d.ii. b for exactly three (1 mark)

d.iii. b for exactly five (1 mark)

Question 4 (10 marks)

2024 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 14\)](#) ↗

Pollutant entering a pond: maximum inflow rate, related rates for the spreading radius, an arctan antiderivative, the limiting area, then a clean-up race

a. Maximum inflow rate and when (1 mark)

b. Rate of increase of the radius at $t = 4$ (3 marks)

c.i. Integral in u (1 mark)

c.ii. V in the required form (1 mark)

d. Limiting surface area (2 marks)

e. Days until the pond is clear (2 marks)

Question 5 (11 marks)

2024 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 17\)](#) ↗

Yacht on a hyperbolic path $3 \sec t \hat{i} + 2 \tan t \hat{j}$: Cartesian form, sketch, speed and its minimum, distance travelled, closest approach to a drone

a. Show the Cartesian equation (1 mark)

b. Sketch with endpoints and direction (2 marks)

c.i. Speed squared in terms of $\sec t$ (1 mark)

c.ii. Time of minimum speed (1 mark)

c.iii. Minimum speed (1 mark)

c.iv. Position at minimum speed (1 mark)

d.i. Integral for the distance (1 mark)

d.ii. Distance to 1 d.p. (1 mark)

e. Shortest drone–yacht distance (2 marks)

Question 6 (10 marks)

2025 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 16\)](#) ↗

Salt-tank mixing problem: reasoning with concentrations, setting up the DE, Euler's method, solving, the limit, a time, then the tap is turned off

a. Does the salt increase? (1 mark)

b. Show the DE (1 mark)

c. Euler's method for $Q(30)$ (2 marks)

d. Solve the DE (3 marks)

e. Limiting quantity (1 mark)

f. Time to reach 100 kg (1 mark)

g. Minutes until concentration is $1/20$ kg/L (1 mark)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide · Exam 1

Questions (technology free)

Question 1 (3 marks) · state average 1.5 / 3 (50%)

2023 VCAA Exam 1 · Question 2

Find b

- ☐ $\arg(z) = 3 \arg(b - i)$
- ☐ $\arg(b - i) = -\frac{\pi}{6}$ (since $-\frac{\pi}{2} < \arg(b - i) < 0$)
- ☐ $b = \sqrt{3}$

Answer / notes: $3 \arg(b - i) = -\frac{\pi}{2} \Rightarrow \tan^{-1}\left(\frac{1}{b}\right) = \frac{\pi}{6}$, so $b = \sqrt{3}$.

State average 1.5 / 3 (50%); 37% scored full marks. Mark distribution: 0: 38% · 1: 14% · 2: 11% · 3: 37%

Your mark: _____ / 3

Question 2 (3 marks) · state average 1.4 / 3 (47%)

2023 VCAA Exam 1 · Question 3

a. Acceleration when $x = 2$

- ☐ Uses $a = v \frac{dv}{dx}$, with $\frac{dv}{dx} = \frac{-7}{(2x-1)^2}$
- ☐ $a = \frac{8}{3} \times \left(-\frac{7}{9}\right) = -\frac{56}{27}$

Answer / notes: $-\frac{56}{27}$ m/s². Many students found $\frac{dv}{dx}$ and stopped: that isn't acceleration.

State average 0.9 / 2 (45%); 35% scored full marks. Mark distribution: 0: 44% · 1: 21% · 2: 35%

Your mark: _____ / 2

b. Limiting velocity as x grows

- ☐ $\frac{3}{2}$

Answer / notes: $v \rightarrow \frac{3}{2}$ m/s.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: _____ / 1

Question 3 (4 marks) · state average 2.1 / 4 (52%)

2023 VCAA Exam 1 · Question 8

Induction proof

- ☐ Base case $n = 1$: $f'(x) = (2x + 1)e^{2x}$ matches
- ☐ Assumes true for $n = k$
- ☐ Differentiates the $n = k$ form: $(2^{k+1}x + (k + 1)2^k)e^{2x}$
- ☐ Concludes true for all n by induction

Answer / notes: Differentiate $(2^kx + k2^{k-1})e^{2x}$ to get $(2^{k+1}x + (k + 1)2^k)e^{2x}$. Lost marks were usually for a missing conclusion or not using the assumption.

State average 2.1 / 4 (52%); 21% scored full marks. Mark distribution: 0: 16% · 1: 17% · 2: 39% · 3: 9% · 4: 21%

Your mark: _____ / 4

Question 4 (6 marks) · state average 3.2 / 6 (53%)

2023 VCAA Exam 1 · Question 10

a. Write $5 - 6 \sin^2 t$ as $\alpha + \beta \cos(2t)$

- ☐ $2 + 3 \cos(2t)$

Answer / notes: $\alpha = 2$, $\beta = 3$

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

Your mark: _____ / 1

b. Show the Cartesian equation of the path

☐ $x - 2 = 3 \cos(2t), y - 1 = 3 \sin(2t)$

☐ Squares and adds to get $(x - 2)^2 + (y - 1)^2 = 9$

Answer / notes: Use $6 \sin t \cos t = 3 \sin(2t)$, then $\cos^2 + \sin^2 = 1$.

State average 1.4 / 2 (70%); 69% scored full marks. Mark distribution: 0: 25% · 1: 6% · 2: 69%

Your mark: _____ / 2

c. Find a from the distance travelled

☐ $a = \frac{\pi}{8}$

Answer / notes: Speed is constant at 6, so $6a = \frac{3\pi}{4}$, $a = \frac{\pi}{8}$.

State average 0.4 / 1 (40%); 36% scored full marks. Mark distribution: 0: 64% · 1: 36%

Your mark: _____ / 1

d. All t with $\tilde{r}(t) \perp \tilde{\dot{r}}(t)$

☐ $\tilde{r} \cdot \tilde{\dot{r}} = 6 \cos(2t) - 12 \sin(2t) = 0$

☐ $t = \frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right) + \frac{k\pi}{2}, k = 0, 1, 2, \dots$

Answer / notes: $\tan(2t) = \frac{1}{2}$. Many students gave only one solution or forgot $t \geq 0$.

State average 0.6 / 2 (30%); 7% scored full marks. Mark distribution: 0: 44% · 1: 49% · 2: 7%

Your mark: _____ / 2

Question 5 (4 marks) · state average 1.9 / 4 (48%)

2024 VCAA Exam 1 · Question 9

a. Is the device activated?

☐ At $x = 0$: $v^2 = 1600 + 336 = 1936$, $v = 44 \geq 44$, so yes

Answer / notes: $v = 44$ km/h, exactly 10% over 40, so it is activated. Many assumed any speed above 40 triggers it.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: _____ / 1

b. Acceleration at $x = 12$

☐ $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

☐ $= -\frac{336}{\pi \sqrt{400 - x^2}}$

☐ $-\frac{21}{\pi}$

Answer / notes: $a = -\frac{336}{16\pi} = -\frac{21}{\pi}$ km/h². Don't forget the $\frac{1}{2}$.

State average 1.6 / 3 (53%); 35% scored full marks. Mark distribution: 0: 29% · 1: 19% · 2: 17% · 3: 35%

Your mark: _____ / 3

Question 6 (3 marks) · state average 0.8 / 3 (27%)

2024 VCAA Exam 1 · Question 10

Values of m

☐ Common normal $\underset{\sim 1}{d} \times \underset{\sim 2}{d} = \underset{\sim}{i} - \underset{\sim}{3j} + \underset{\sim}{5k}$

☐ Distance $\frac{|5m+4|}{\sqrt{35}} = \frac{14}{\sqrt{35}}$

☐ $m = 2$ or $m = -\frac{18}{5}$

Answer / notes: Project the vector between points on each line onto the common normal. $m = 2, -\frac{18}{5}$. The state averaged 0.8/3.

State average 0.8 / 3 (27%); 14% scored full marks. Mark distribution: 0: 59% · 1: 14% · 2: 13% · 3: 14%

Your mark: _____ / 3

Question 7 (5 marks) · state average 2.7 / 5 (54%)

2025 VCAA Exam 1 · Question 4

a. Show $E(T) = \frac{1}{2}$

☐ $E(T) = \frac{3}{2 \log_e 2} \int_0^1 \frac{t}{(t+1)(2-t)} dt$

☐ Partial fractions: $\frac{t}{(t+1)(2-t)} = -\frac{1}{3(t+1)} + \frac{2}{3(2-t)}$

☐ Integral $= \frac{1}{3} \log_e 2$, so $E(T) = \frac{1}{2}$

Answer / notes: Partial fractions are the key step. Many lost a mark by not showing enough working for a “show that”.

State average 1.5 / 3 (50%); 37% scored full marks. Mark distribution: 0: 32% · 1: 23% · 2: 9% · 3: 37%

Your mark: _____ / 3

b. $\Pr(0.44 < \bar{T} < 0.5)$

☐ $\text{sd}(\bar{T}) = \frac{0.3}{5} = 0.06$

☐ $\Pr(-1 < Z < 0) = 0.34$

Answer / notes: $0.84 - 0.5 = 0.34$.

State average 1.2 / 2 (60%); 49% scored full marks. Mark distribution: 0: 34% · 1: 17% · 2: 49%

Your mark: _____ / 2

Question 8 (4 marks) · state average 2.1 / 4 (52%)

2025 VCAA Exam 1 · Question 5

a. Show $c = -4$

☐ Equates $\tilde{j}$ components at $t = 1$: $-1 = 3 + c$

Answer / notes: $-1 = 2 + c + 1$, so $c = -4$.

State average 0.9 / 1 (90%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: _____ / 1

b. Two possible values of a

☐ From the $\tilde{i}$ components, $b = a - 1$; velocities $(3 + 2a)\tilde{i}$ and $(b + 2)\tilde{i} + \tilde{j}$

☐ Dot product zero: $(3 + 2a)(a + 1) = 0$, so $a = -\frac{3}{2}$ or $a = -1$

Answer / notes: $a = -1$ or $a = -\frac{3}{2}$.

State average 1.0 / 2 (50%); 30% scored full marks. Mark distribution: 0: 33% · 1: 37% · 2: 30%

Your mark: _____ / 2

c. Values of a and b

☐ $|6 + 2a| = 4$ combined with part b: $a = -1$, $b = -2$

Answer / notes: $a = -1$, $b = -2$. Listing extra pairs (e.g. $a = -5$) lost the mark.

State average 0.2 / 1 (20%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: _____ / 1

Question 9 (6 marks) · state average 2.7 / 6 (45%)

2025 VCAA Exam 1 · Question 9

a. Show $f(x) = -x - 1 + \frac{1}{x+1}$

☐ Factorises: $\frac{x(x+2)(x-1)}{-(x-1)(x+1)} = -\frac{x(x+2)}{x+1}$

☐ Rewrites $-\frac{(x+1)^2 - 1}{x+1}$ to get the form

Answer / notes: Cancel the $(x - 1)$ factor first, then divide. Check every line actually leads to the given form.

State average 1.2 / 2 (60%); 45% scored full marks. Mark distribution: 0: 22% · 1: 33% · 2: 45%

Your mark: _____ / 2

b. k for continuity at $x = 1$

☐ $k = -\frac{3}{2}$

Answer / notes: $k = -1 - 1 + \frac{1}{2} = -\frac{3}{2}$.

State average 0.5 / 1 (50%); 46% scored full marks. Mark distribution: 0: 54% · 1: 46%

Your mark: _____ / 1

c. Sketch, labelling asymptotes

- ☐ Asymptotes $x = -1$ and $y = -x - 1$ labelled
- ☐ Correct branches through $(0, 0)$ and $(-2, 0)$
- ☐ Open circle at $(1, -\frac{3}{2})$

Answer / notes: No asymptote at $x = 1$: it's a hole. Many drew one there or forgot the open circle.

State average 1.0 / 3 (33%); 16% scored full marks. Mark distribution: 0: 50% · 1: 16% · 2: 18% · 3: 16%

Your mark: _____ / 3

Marking guide · Exam 2

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	E	$\arg z \in (0, \frac{\pi}{2})$ and $3 \arg z$ must be π , so $z = 4 \operatorname{cis} \frac{\pi}{3}$ and $z^2 = 16 \operatorname{cis} \frac{2\pi}{3}$. That equals $-4\bar{z}$. <i>Most popular wrong answer: B (20%).</i>	34% correct A 15% · B 20% · C 13% · D 18% · E 34%
☐	2	A	Volume $8000 - 5t$; outflow $20 \cdot \frac{Q}{8000-5t} = \frac{4Q}{1600-t}$, so $\frac{dQ}{dt} = -\frac{4Q}{1600-t} = \frac{4Q}{t-1600}$. The popular wrong answer dropped the minus sign. <i>Most popular wrong answer: E (29%).</i>	37% correct A 37% · B 6% · C 13% · D 14% · E 29%
☐	3	A	Parts with $u = (1-x)^n$: $[(1-x)^n e^x]_0^1 + n \int_0^1 (1-x)^{n-1} e^x dx = -1 + nI_{n-1}$. <i>Most popular wrong answer: E (25%).</i>	33% correct A 33% · B 10% · C 23% · D 7% · E 25%
☐	4	E	Use $x = \cos y$: $S = 2\pi \int_0^{\pi/2} \cos y \sqrt{1 + \sin^2 y} dy$, then $u = \sin y$ changes the terminals to 0 and 1. <i>Most popular wrong answer: C (22%).</i>	46% correct A 8% · B 6% · C 22% · D 18% · E 46%
☐	5	E	The phone starts with the balloon's upward velocity 2.5 m/s: $-80 = 2.5t - 4.9t^2$, so $t \approx 4.30$ s. Ignoring the initial velocity gives a wrong option. <i>Most popular wrong answer: C (22%).</i>	45% correct A 4% · B 10% · C 22% · D 19% · E 45%
☐	6	B	$\tilde{n}$ is perpendicular to both $\tilde{i} \pm \tilde{j}$, so $\tilde{n} = \pm \tilde{k}$ and $ \tilde{c} \cdot \tilde{n} = 3$. <i>Most popular wrong answer: E (19%).</i>	48% correct A 7% · B 48% · C 16% · D 9% · E 19%
☐	7	D	$ \tilde{a} + \tilde{b} ^2 = 2 + 2\tilde{a} \cdot \tilde{b} = 1$, so $\tilde{a} \cdot \tilde{b} = -\frac{1}{2}$ and $ \tilde{a} - \tilde{b} ^2 = 3$. <i>Most popular wrong answer: C (32%).</i>	18% correct A 22% · B 24% · C 32% · D 18% · E 3%
☐	8	D	Max height ≈ 13 m reached from a 1.5 m start, so up 11.5 m then down 13 m: about 24.5 m. Doubling the max height ignores the launch height. <i>Most popular wrong answer: C (23%).</i>	32% correct A 8% · B 13% · C 23% · D 32% · E 23%
☐	9	A	$\frac{(x+5)(x-2)}{x-2} = x+5$, and the defined value at $x=2$ is exactly $2+5$, so the function is continuous. Many thought there was still a discontinuity. <i>Most popular wrong answer: D (37%).</i>	48% correct A 48% · B 10% · C 4% · D 37%
☐	10	A	$\cos x = \sqrt{1-a^2}$ (fourth quadrant), and $\frac{\pi}{2} \in (\frac{3\pi}{4}, \pi)$ so $\cos \frac{\pi}{2} < 0$: $-\sqrt{\frac{1+\sqrt{1-a^2}}{2}}$. <i>Most popular wrong answer: C (38%).</i>	27% correct A 27% · B 23% · C 38% · D 10%
☐	11	A	$\dot{x}^2 + \dot{y}^2 = 2 - 2\cos t = 4\sin^2 \frac{t}{2}$, and $\sin \frac{t}{2} \geq 0$ on $[0, 2\pi]$, so the integrand is $2\sin \frac{t}{2}$. <i>Most popular wrong answer: C (29%).</i>	45% correct A 45% · B 11% · C 29% · D 14%
☐	12	B	$\frac{\tilde{r} \cdot \tilde{s}}{ \tilde{s} } = -2$, so $\tilde{r} \cdot \tilde{s} = -18$ and the scalar resolute of $\tilde{s}$ on $\tilde{r}$ is $\frac{-18}{9} = -2$. Watch which vector you resolve onto. <i>Most popular wrong answer: C (33%).</i>	36% correct A 14% · B 36% · C 33% · D 17%
☐	13	A	$y = 1 - 2x^2$. It starts at $(0, 1)$, reaches $(1, -1)$ at $t = \frac{\pi}{2}$, turns, reaches $(-1, -1)$ at $\frac{3\pi}{2}$, back to $(0, 1)$ at 2π . The common wrong choice had period π . <i>Most popular wrong answer: C (44%).</i>	36% correct A 36% · B 12% · C 44% · D 8%
☐	14	A	$\tilde{v}_1 \cdot \tilde{r} = \cos^2 t - \sin^2 t + \cos 2t = 2\cos 2t$, zero only at $t = \frac{\pi}{4}$ in $(0, \frac{\pi}{2})$. <i>Most popular wrong answer: B (30%).</i>	43% correct A 43% · B 30% · C 19% · D 7%
☐	15	B	Total $= 0.7(M_1 + \dots + M_4) \sim N(560, 10.5^2)$. $\Pr(T > 570) = \Pr(Z > 0.952) \approx 0.1705$. Using $0.7 \times 4M$ gives the wrong sd. <i>Most popular wrong answer: C (25%).</i>	48% correct A 9% · B 48% · C 25% · D 17%

✓	Q	Ans	Why	How Victoria went
☐	16	D	$f(x) = x^4 - x$ has $f''(x) = 12x^2$: zero at 0 but no sign change, so no inflection. The other options do have inflections at 0. <i>Most popular wrong answer: C (40%).</i>	48% correct A 5% · B 6% · C 40% · D 48%
☐	17	C	$2\pi \int e^{kt} \sqrt{k^2 + k^2 e^{2kt}} dt$, then $u = e^{kt}$, $du = ke^{kt} dt$: $2\pi \int_{e^{ka}}^{e^{kb}} \sqrt{1+u^2} du$. <i>Most popular wrong answer: D (29%).</i>	49% correct A 8% · B 15% · C 49% · D 29%
☐	18	D	$v_m^2 = u^2 + 2a\frac{s}{2}$ and $v^2 = u^2 + 2as$, so $v_m = \sqrt{\frac{u^2+v^2}{2}}$. The average $\frac{u+v}{2}$ is the velocity at the midpoint in time. <i>Most popular wrong answer: A (33%).</i>	33% correct A 33% · B 13% · C 20% · D 33%
☐	19	B	$ a b \cos \theta = a b \sin \theta$, so $\tan \theta = 1$, $\theta = \frac{\pi}{4}$. <i>Most popular wrong answer: C (30%).</i>	48% correct A 17% · B 48% · C 30% · D 5%
☐	20	D	From line 1, $x = 4$ gives $\lambda = 2$, so $r = 5$, $t = 5$. From line 2, $\mu = 3$, so $s = 8$. <i>Most popular wrong answer: B (23%).</i>	49% correct A 9% · B 23% · C 19% · D 49%

Multiple-choice score: _____ / 20

Section B – Extended-response questions

Question 1 (10 marks) · state average 4.2 / 10 (42%)

2023 VCAA Exam 2 · Section B · Question 2

a. Verify $w = \text{cis } \frac{2\pi}{7}$ is a root

☐ $w^7 = \text{cis}(2\pi) = 1$ by De Moivre

Answer / notes: $w^7 = \text{cis}(2\pi) = 1$.

State average 0.6 / 1 (60%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

b. Other roots in polar form

☐ 1, $\text{cis}(\pm \frac{2\pi}{7})$ (excluding w), $\text{cis}(\pm \frac{4\pi}{7})$, $\text{cis}(\pm \frac{6\pi}{7})$

Answer / notes: $\text{cis}(0)$, $\text{cis}(-\frac{2\pi}{7})$, $\text{cis}(\pm \frac{4\pi}{7})$, $\text{cis}(\pm \frac{6\pi}{7})$. Omitting 1 was common.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

c. Plot all seven roots

☐ Seven points on the unit circle

☐ Spaced $\frac{2\pi}{7}$ apart, starting at 1

Answer / notes: The diagram's sectors are not sevenths: place the points by angle, not by the printed lines.

State average 1.2 / 2 (60%); 53% scored full marks. Mark distribution: 0: 35% · 1: 12% · 2: 53%

Your mark: _____ / 2

d.i. Sketch the ray from 1 through w

☐ Ray starting at $z = 1$ (open circle) through $\text{cis } \frac{2\pi}{7}$

Answer / notes: Start at 1 and extend beyond w .

State average 0.4 / 1 (40%); 42% scored full marks. Mark distribution: 0: 58% · 1: 42%

Your mark: _____ / 1

d.ii. Equation of the ray

☐ $\text{Arg}(z - 1) = \frac{9\pi}{14}$

Answer / notes: The chord from 1 to $\text{cis } \phi$ makes angle $\frac{\pi}{2} + \frac{\phi}{2}$, so $\frac{9\pi}{14}$.

State average 0.2 / 1 (20%); 18% scored full marks. Mark distribution: 0: 82% · 1: 18%

Your mark: _____ / 1

e. Verify the factorised form of $z^7 - 1$

☐ Expands $(z - 1)(z^6 + \dots + 1)$ to $z^7 - 1$

Answer / notes: Expand and watch the terms telescope.

State average 0.5 / 1 (50%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: _____ / 1

f.i. Write $\text{cis } \frac{2\pi}{7} + \text{cis } \frac{12\pi}{7}$ as $A \cos(B\pi)$

☐ $2 \cos\left(\frac{2\pi}{7}\right)$

Answer / notes: $\text{cis } \frac{12\pi}{7} = \text{cis}\left(-\frac{2\pi}{7}\right)$, so the sum is $2 \cos \frac{2\pi}{7}$.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: _____ / 1

f.ii. Show the cosine sum is $-1/2$

☐ Since $w \neq 1$, $w^6 + w^5 + \dots + w + 1 = 0$

☐ Pairs $w^k + w^{7-k} = 2 \cos \frac{2k\pi}{7}$, so $1 + 2(\text{sum}) = 0$

Answer / notes: Average 0.2/2: you need the logical chain, not just the final line.

State average 0.2 / 2 (10%); 7% scored full marks. Mark distribution: 0: 85% · 1: 8% · 2: 7%

Your mark: _____ / 2

Question 2 (11 marks) · state average 6.0 / 11 (55%)

2023 VCAA Exam 2 · Section B · Question 5

a. Vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and show the area is 1.5

☐ $\overrightarrow{AB} = \underset{\sim}{j} + \underset{\sim}{k}$, $\overrightarrow{AC} = 2\underset{\sim}{i} + \underset{\sim}{j} + 2\underset{\sim}{k}$

☐ $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2}|(1, 2, -2)| = \frac{3}{2}$

Answer / notes: Show the cross product and its magnitude explicitly.

State average 1.6 / 2 (80%); 63% scored full marks. Mark distribution: 0: 8% · 1: 29% · 2: 63%

Your mark: _____ / 2

b. Shortest distance from B to AC

☐ Uses area $= \frac{1}{2}|AC| \times d$ (or a vector resolute)

☐ $d = 1$

Answer / notes: $\frac{2 \times 1.5}{|AC|} = \frac{3}{3} = 1$.

State average 0.7 / 2 (35%); 29% scored full marks. Mark distribution: 0: 60% · 1: 11% · 2: 29%

Your mark: _____ / 2

c. Acute angle between the line and plane ψ

☐ $\sin \theta = \frac{|\underset{\sim}{d} \cdot \underset{\sim}{n}|}{|\underset{\sim}{d}| |\underset{\sim}{n}|} = \frac{4}{9}$

☐ $\theta \approx 26^\circ$

Answer / notes: 26° . Many stopped at the angle with the normal (64°).

State average 1.2 / 2 (60%); 50% scored full marks. Mark distribution: 0: 29% · 1: 21% · 2: 50%

Your mark: _____ / 2

d. Parametric equation of L

☐ $x = 2t$, $y = -2t$, $z = -t$

Answer / notes: Direction is the normal to ψ , through the origin.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: _____ / 1

e. Shortest distance from the origin to ψ

☐ $\frac{|-18|}{\sqrt{4+4+1}}$

☐ 6

Answer / notes: Distance is positive: use the absolute value properly.

State average 1.2 / 2 (60%); 55% scored full marks. Mark distribution: 0: 37% · 1: 7% · 2: 55%

Your mark: _____ / 2

f. Coordinates of D

☐ Substitutes L into ψ : $9t = -18$

☐ $D(-4, 4, 2)$

Answer / notes: $t = -2$.

State average 0.8 / 2 (40%); 34% scored full marks. Mark distribution: 0: 53% · 1: 13% · 2: 34%

Your mark: _____ / 2

Question 3 (10 marks) · state average 5.6 / 10 (56%)

2024 VCAA Exam 2 · Section B · Question 1

a. Sketch with asymptotes and stationary points

☐ Vertical asymptotes $x = \pm 1$ labelled

☐ Stationary points $(0, 1)$ and $(\pm\sqrt{2}, -3)$ labelled

☐ Correct shape: U-shaped middle branch, outer branches turning at $(\pm\sqrt{2}, -3)$ and heading down

Answer / notes: $f(x) = -x^2 + \frac{1}{1-x^2}$. Sketch on CAS first, matching the given scale.

State average 2.0 / 3 (67%); 17% scored full marks. Mark distribution: 0: 2% · 1: 16% · 2: 65% · 3: 17%

Your mark: _____ / 3

b.i. Integral in y for the volume

☐ Solves for x^2 : $x^2 = \frac{1-y+\sqrt{(y-1)(y+3)}}{2}$

☐ $V = \pi \int_1^6 \frac{1-y+\sqrt{y^2+2y-3}}{2} dy$

Answer / notes: Write the actual expression for x^2 inside the integral, not a generic formula.

State average 1.4 / 2 (70%); 56% scored full marks. Mark distribution: 0: 22% · 1: 22% · 2: 56%

Your mark: _____ / 2

b.ii. Volume to 1 d.p.

☐ 11.2

Answer / notes: ≈ 11.2 .

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: _____ / 1

c. b for no asymptotes

☐ $b = -1$

Answer / notes: $g(x) = -x^2 - 1 + \frac{b+1}{1-x^2}$, so $b = -1$.

State average 0.7 / 1 (70%); 72% scored full marks. Mark distribution: 0: 28% · 1: 72%

Your mark: _____ / 1

d.i. b for exactly one stationary point

☐ $b \leq -1$

Answer / notes: $x = 0$ is always stationary; you need $(x^2 - 1)^2 = b + 1$ to have no valid solutions.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: _____ / 1

d.ii. b for exactly three

☐ $b \geq 0$

Answer / notes: Only $x^2 = 1 + \sqrt{b+1}$ gives new points. Include the equality.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: _____ / 1

d.iii. b for exactly five

☐ $-1 < b < 0$

Answer / notes: Both $x^2 = 1 \pm \sqrt{b+1}$ positive and not 1.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: _____ / 1

Question 4 (10 marks) · state average 4.4 / 10 (44%)

2024 VCAA Exam 2 · Section B · Question 3

a. Maximum inflow rate and when

☐ $0.05 \text{ m}^3/\text{day}$ at $t = 2$

Answer / notes: Set the derivative of $\frac{8t}{240+5t^4}$ to zero: $t^4 = 16$.

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

Your mark: _____ / 1

b. Rate of increase of the radius at $t = 4$

☐ $V = 0.001\pi r^2$, so $\frac{dV}{dt} = 0.002\pi r \frac{dr}{dt}$

☐ $\frac{dV}{dt} = \frac{32}{1520}$ at $t = 4$

☐ $\frac{dr}{dt} \approx 0.51 \text{ m/day}$

Answer / notes: Convert the 1 mm depth to metres.

State average 1.6 / 3 (53%); 37% scored full marks. Mark distribution: 0: 35% · 1: 11% · 2: 17% · 3: 37%

Your mark: _____ / 3

c.i. Integral in u

☐ $\int \frac{4}{\sqrt{5}(240+u^2)} du$

Answer / notes: $du = 2\sqrt{5}t dt$, so $8t dt = \frac{4}{\sqrt{5}} du$.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

c.ii. V in the required form

☐ $V = \frac{1}{5\sqrt{3}} \arctan\left(\frac{t^2}{4\sqrt{3}}\right)$

Answer / notes: $a = 5, b = 3, c = 2, d = 4$.

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

d. Limiting surface area

☐ $V \rightarrow \frac{\pi}{10\sqrt{3}}$

☐ $\text{Area} = \frac{V}{0.001} \approx 181.38 \text{ m}^2$

Answer / notes: Area = volume ÷ depth. Many skipped this.

State average 0.5 / 2 (25%); 17% scored full marks. Mark distribution: 0: 73% · 1: 10% · 2: 17%

Your mark: _____ / 2

e. Days until the pond is clear

☐ Solves $V(t) = 0.05(t - 5)$: $t \approx 8.40$

☐ 3.4 days after clean-up starts

Answer / notes: The common error was ignoring the 5-day delay.

State average 0.3 / 2 (15%); 4% scored full marks. Mark distribution: 0: 79% · 1: 17% · 2: 4%

Your mark: _____ / 2

Question 5 (11 marks) · state average 5.6 / 11 (51%)

2024 VCAA Exam 2 · Section B · Question 4

a. Show the Cartesian equation

☐ Uses $\sec^2 t - \tan^2 t = 1$

Answer / notes: $\frac{x^2}{9} - \frac{y^2}{4} = \sec^2 t - \tan^2 t = 1$.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: _____ / 1

b. Sketch with endpoints and direction

☐ Left branch from $(-6, -2\sqrt{3})$ to $(-6, 2\sqrt{3})$ through $(-3, 0)$

☐ Arrow showing motion upwards (y increasing)

Answer / notes: Negative signs on endpoints were often lost.

State average 1.0 / 2 (50%); 41% scored full marks. Mark distribution: 0: 36% · 1: 22% · 2: 41%

Your mark: _____ / 2

c.i. Speed squared in terms of $\sec t$

☐ $13 \sec^4 t - 9 \sec^2 t$

Answer / notes: $9 \sec^2 t \tan^2 t + 4 \sec^4 t$, then $\tan^2 = \sec^2 - 1$.

State average 0.3 / 1 (30%); 23% scored full marks. Mark distribution: 0: 77% · 1: 23%

Your mark: _____ / 1

c.ii. Time of minimum speed

☐ $t = \pi$

Answer / notes: The derivative has a factor $\tan t$, zero at $t = \pi$.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

c.iii. Minimum speed

☐ 2 m/min

Answer / notes: Take the square root of speed squared.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: _____ / 1

c.iv. Position at minimum speed

☐ $(-3, 0)$

Answer / notes: $\tilde{r}(\pi) = -3i$: the vertex of the branch. Watch the negative sign.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: _____ / 1

d.i. Integral for the distance

☐ $\int_{2\pi/3}^{4\pi/3} \sqrt{13 \sec^4 t - 9 \sec^2 t} dt$

Answer / notes: Use brackets carefully.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

d.ii. Distance to 1 d.p.

☐ 9.4 m

Answer / notes: ≈ 9.4 m.

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: _____ / 1

e. Shortest drone–yacht distance

☐ $|r_{\tilde{D}}(t) - r_{\tilde{Y}}(t)|$ set up

☐ Minimum ≈ 11.1 m

Answer / notes: Both use the same t . Minimise numerically over the yacht's domain.

State average 0.6 / 2 (30%); 24% scored full marks. Mark distribution: 0: 65% · 1: 10% · 2: 24%

Your mark: _____ / 2

Question 6 (10 marks) · state average 5.5 / 10 (55%)

2025 VCAA Exam 2 · Section B · Question 3

a. Does the salt increase?

☐ Initial concentration $\frac{5}{3000} = \frac{1}{600}$ kg/L is less than the inflow 0.1 kg/L, so it increases

Answer / notes: Quote both concentrations. Only 20% did.

State average 0.2 / 1 (20%); 20.43% scored full marks. Mark distribution: 0: 79.57% · 1: 20.43%

Your mark: _____ / 1

b. Show the DE

☐ Rate in = 2, rate out = $\frac{20Q}{3000}$, so $\frac{dQ}{dt} = \frac{300-Q}{150}$

Answer / notes: Show rate in – rate out.

State average 0.6 / 1 (61%); 61.55% scored full marks. Mark distribution: 0: 38.45% · 1: 61.55%

Your mark: _____ / 1

c. Euler's method for $Q(30)$

☐ $Q(15) = 5 + 15 \times \frac{295}{150} = 34.5$

☐ $Q(30) = 34.5 + 15 \times \frac{265.5}{150} = 61.05$

Answer / notes: Show the intermediate value.

State average 0.9 / 2 (46%); 38.93% scored full marks. Mark distribution: 0: 46.55% · 1: 14.52% · 2: 38.93%

Your mark: _____ / 2

d. Solve the DE

☐ $\int \frac{150}{300-Q} dQ = \int dt$

☐ $-150 \log_e(300 - Q) = t + c$

☐ $Q = 300 - 295e^{-t/150}$

Answer / notes: Use the initial condition $Q(0) = 5$.

State average 2.2 / 3 (75%); 59.3% scored full marks. Mark distribution: 0: 11.24% · 1: 12.17% · 2: 17.3% · 3: 59.3%

Your mark: _____ / 3

e. Limiting quantity

☐ 300 kg

Answer / notes: $e^{-t/150} \rightarrow 0$, so $Q \rightarrow 300$ kg (concentration matches the inflow).

State average 0.8 / 1 (79%); 79.46% scored full marks. Mark distribution: 0: 20.54% · 1: 79.46%

Your mark: _____ / 1

f. Time to reach 100 kg

☐ $t = 150 \log_e \frac{59}{40}$

Answer / notes: Exact form required.

State average 0.6 / 1 (60%); 60.7% scored full marks. Mark distribution: 0: 39.3% · 1: 60.7%

Your mark: _____ / 1

g. Minutes until concentration is 1/20 kg/L

☐ 50

Answer / notes: $\frac{100+2t}{3000+20t} = \frac{1}{20}$. The common wrong answer was 25.

State average 0.2 / 1 (18%); 18.05% scored full marks. Mark distribution: 0: 81.95% · 1: 18.05%

Your mark: _____ / 1

Your result

Total: _____ / 120 State average on these questions: **58 / 120 (48%)**

Topics to revise:

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